Is $\|S\|_{L^\infty} \le C\|\omega\|_{L^\infty}$? A deep check of `PeriodicStrainLinfBound`

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Lean record: `PeriodicNSContinuation/{StrainLinf,StrainLinfFamily,SineSumBound}.lean` (Apache-2.0), repository `github.com/BenFrohman/periodic-ns-classical-continuation`, local commit `029a9b9` (not yet pushed; the Lean files are attached to this record).
This document does not claim a solution of the Navier–Stokes Millennium Problem.

Abstract

Is there one constant C with $\|Su(x)\| \le C\|\omega u\|_{L^\infty}$ for every smooth, divergence-free, $\mathbb{Z}^3$-periodic field u? We prove in Lean 4/Mathlib, with no `sorry` and only the standard axioms, that there is not (`not_staticStrainLinfBound`). An explicit family u_K of divergence-free trigonometric polynomials (K odd) has the Fourier identity $S_{11}(0) = \frac{16}{\pi^2}\sum_{k,l \text{ odd}\le K}(k^2+l^2)^{-1}$ (proved), the uniform vorticity bound $\|\omega_K\|_\infty \le 36$ (proved, via a formalized Gibbs/Fejér–Jackson-type bound $|\sum_{k\le n}\sin(kx)/k| \le 3$; the sharper $(6\,\mathrm{Si}(\pi)/\pi)^2 \approx 12.5$ is classical), and $\|S_K\|_\infty \ge \frac{1}{\pi^2}\log K$ for all odd K (proved). The sharp growth $\frac{2}{\pi}\log K$ is numerical only. The solution-level hypothesis `PeriodicStrainLinfBound` is false as stated only modulo classical local existence, which is not formalized. We record the conditional results that do give the bound.

1 Literature

Biot–Savart gives $\nabla u = \mathcal{R}\omega$ with $\mathcal{R}$ a matrix of Calderón–Zygmund operators (compositions of Riesz transforms). These are bounded on L^p, $1 < p < \infty$, and from L^∞ to BMO, but not on L^∞. For bounded vorticity the velocity is only log-Lipschitz (Yudovich; see Majda–Bertozzi, *Vorticity and Incompressible Flow*, Ch. 8). The standard sharp example is the Bahouri–Chemin steady state $\omega = \operatorname{sgn} x \operatorname{sgn} y$ on $\mathbb{T}^2$, whose velocity gradient diverges logarithmically at the origin (Bahouri–Chemin 1994; used by Denisov for double-exponential growth of $\nabla\omega$ in 2D Euler, and by Elgindi–Jeong in 3D symmetric settings). The quantitative substitute is the log inequality of Beale–Kato–Majda and Kozono–Taniuchi, $\|\nabla u\|_\infty \le C(1 + \|\omega\|_{\mathrm{BMO}}(1 + \log^+\|u\|_{H^s}))$.

2 An explicit periodic counterexample family

Let $f_K(x) = \frac{4}{\pi}\sum_{k \text{ odd}\le K}\sin(kx)/k$, the Dirichlet partial sum of $\operatorname{sgn}(\sin x)$. On $\mathbb{T}^3$ put

$$\omega_K = f_K(x)f_K(y)\,e_z, \qquad \Delta\psi_K = f_K f_K, \qquad u_K = (-\partial_y\psi_K,\ \partial_x\psi_K,\ 0).$$

Then u_K is a divergence-free trigonometric polynomial with $\nabla\times u_K = \omega_K$. With $\omega_K = \sum A_{kl}\sin kx\sin ly$, $A_{kl} = 16/(\pi^2 kl)$, we have $\psi_K = -\sum A_{kl}\sin kx\sin ly/(k^2+l^2)$ and

$$S_{11}(0) = -\partial_{xy}\psi_K(0) = \frac{16}{\pi^2}\sum_{k,l\ \mathrm{odd}\le K}\frac{1}{k^2+l^2}, \qquad S_{12}(0) = 0,$$

so $\|S_K(0)\|_{op} = S_{11}(0)$.

Theorem 1 (proved in Lean, commit `029a9b9`)**.** *`uK_initialData`: u_K is smooth, divergence-free and $\mathbb{Z}^3$-periodic. `vort_eq`: $\omega_K = f_K(x)f_K(y)e_z$. `strain00_origin`: the Fourier identity above, $\langle e_1, S_K(0)e_1\rangle = \frac{16}{\pi^2}$ `bcSum`n, $n = (K+1)/2$. `strain_norm_origin_ge`: $\|S_K(0)\| \ge \frac{16}{\pi^2}$ `bcSum`n. `bcSum_ge_log`, `originStrain_ge_log`: `bcSum`$n \ge \frac{1}{8}\log(n+1)$ and hence $\|S_K(0)\| \ge \frac{1}{\pi^2}\log K$ for all odd K. `bcSum_unbounded`: divergence.*

Vorticity bound. Writing $f_K(x) = g_K(x) - \frac{1}{2}g_{\lfloor K/2\rfloor}(2x)$ with $g_n(x) = \frac{4}{\pi}\sum_{k\le n}\sin(kx)/k$, Lean proves $|\sum_{k\le n}\sin(kx)/k| \le 3$ (`SineSumBound.abs_g_le_three`, via Abel summation and the Dirichlet kernel), hence $|\sum_{k\ \mathrm{odd}\le K}\sin(kx)/k| \le \frac{9}{2}$ (`abs_odd_sum_le`), $|f_K| \le 18/\pi < 6$ (`abs_F_le`) and $\|\omega_K\|_\infty \le 36$ (`norm_vort_le`, `vorticity_sup_norm_uK_le`). Classically, the sharp bound $|\sum_{k\le n}\sin(kx)/k| \le \mathrm{Si}(\pi) \approx 1.852$ (Fejér–Jackson–Gronwall, not formalized) gives $\sup|f_K| \le \frac{6}{\pi}\mathrm{Si}(\pi)$, so $\|\omega_K\|_\infty \le (6\,\mathrm{Si}(\pi)/\pi)^2 \approx 12.5$ for all K. Numerically $\sup|f_K|^2$ decreases in K: 1.621 ($K=1$), 1.441 ($K=3$), 1.412 ($K=5$), 1.4024 ($K=7$), so $\|\omega_K\|_\infty \le 1.403$ for all odd $K \ge 7$ (and $\le 16/\pi^2$ for all K). The limit is $(\frac{2}{\pi}\mathrm{Si}\,\pi)^2 \approx 1.3900$ (Gibbs constant squared).

Strain growth. $\|S_K\|_\infty \ge S_{11}(0) \to \infty$. The asymptotic $S_{11}(0) \sim \frac{2}{\pi}\log K$ is classical (the lattice sum $\sum_{k,l\ \mathrm{odd}\le K}(k^2+l^2)^{-1} \sim \frac{\pi}{8}\log K$) and is confirmed numerically below. Lean proves the non-sharp $\|S_K\|_\infty \ge \frac{1}{\pi^2}\log K$ for all odd K; the sharp constant $\frac{2}{\pi}$ is numerical only. The ratio $\|S\|_\infty/\|\omega\|_\infty$ is invariant under $u \mapsto \lambda u(\mu\,\cdot)$, so no rescaling removes the divergence.

Numerics (complete; exact sums at the origin, grid sup of $\|S\|_{op}$ on an $8K\times 8K$ grid up to $K = 255$).

K	$\|\omega_K\|_\infty$	$S_{11}(0)$	$\sup_x\|S\|_{op}$	$S_{11}(0) - \frac{2}{\pi}\log K$
7	1.4024	1.6585	1.6585	0.4196
15	1.3931	2.0977	2.0977	0.3737
31	1.3908	2.5385	2.5385	0.3523
63	1.3902	2.9796	2.9796	0.3420
127	1.3900	3.4209	3.4209	0.3370
255	1.3900	3.8621	3.8621	0.3345
511	1.3900	4.3034	–	0.3332
1023	1.3900	4.7447	–	0.3326

K	t	M	$\|S\|_\infty$	ratio
31	0.00	1.3908	2.5385	1.8252
31	0.20	1.0757	2.2023	2.0473
31	0.40	1.0199	1.9882	1.9494
31	0.60	1.0075	1.8391	1.8253
31	0.80	1.0042	1.7235	1.7163
31	1.00	1.0023	1.6293	1.6255
127	0.00	1.3899	3.4209	2.4612
127	0.20	1.0000	2.2628	2.2628
127	0.40	1.0000	2.0156	2.0156
127	0.60	1.0000	1.8563	1.8563
127	0.80	1.0000	1.7350	1.7350
127	1.00	1.0000	1.6371	1.6371

The first table shows $\|\omega_K\|_\infty \to 1.3900 = (\frac{2}{\pi}\operatorname{Si}\pi)^2$, the sup of $\|S\|$ attained at the origin, and $S_{11}(0) - \frac{2}{\pi}\log K \to\approx 0.332$: numerically consistent with the classical asymptotic $\frac{2}{\pi}\log K$ (Lean proves $\frac{1}{\pi^2}\log K$). The second table evolves the $K = 31$ and $K = 127$ data by 2D Navier–Stokes ($= z$-independent 3D NS), $\nu = 0.005$, to $t = 1$: the ratio is largest near $t = 0$ and is then quickly smoothed (to ≈ 1.63 by $t = 1$).

3 Conditional results that do give the bound

- **Dini continuity** (classical CZ theory): if $\omega(t)$ has modulus of continuity ϖ with $\int_0^1 \varpi(r)\,dr/r < \infty$, then $\|\nabla u\|_\infty \le C\big(\|\omega\|_\infty + \int_0^1 \varpi(r)\frac{dr}{r} + \|\omega\|_{L^2}\big)$. This includes Hölder and log-Hölder moduli $\varpi(r) \lesssim |\log r|^{-a}$, $a > 1$. The constant depends on the modulus, not on $\|\omega\|_\infty$ alone; this is a different statement.
- **Log inequality** (BKM, Kozono–Taniuchi): bounded by $\|\omega\|_\infty$ or $\|\omega\|_{\mathrm{BMO}}$ times log of a higher norm. This is the right replacement and is what Section 2 of the cubic-source note now uses.
- **Direction regularity** (Constantin–Fefferman): Lipschitz (or $\frac{1}{2}$-Hölder, Beirão da Veiga–Berselli) continuity of $\xi = \omega/|\omega|$ on the high-vorticity set controls the stretching $\xi \cdot S\xi$ through a local, cancelling kernel. It does not give $\|S\|_\infty \le C\|\omega\|_\infty$; the family above is two-dimensional, with constant direction $\xi = e_z$, and still violates the bound.

4 The static bound is refuted in Lean

Definition 1 (`StaticStrainLinfBound`)**.** $\exists C\ \forall u$ smooth, divergence-free, $\mathbb{Z}^3$-periodic, $\forall x$: $\|Su(x)\| \le C\,\|\omega u\|_{L^\infty}$.

Theorem 2 (`not_staticStrainLinfBound`, proved in Lean, no caveat)**.** $\neg$ *`StaticStrainLinfBound`.*

It involves no time, no Navier–Stokes equation and no local existence: $\|\omega_K\|_\infty \le 36$ while $\|S_K(0)\| \to \infty$.

5 Could it hold for Navier–Stokes solutions?

1. **Uniform constant: false as stated, modulo classical local existence (not formalized).** `PeriodicStrainLinfBound` quantifies over all smooth periodic data with one C, at all

$t \geq 0$ including $t = 0$. Each u_K is smooth periodic divergence-free data with a smooth local solution (classical local well-posedness; it is in fact global, being 2D). At $t = 0$ the bound fails as $K \to \infty$. The static inequality at $t = 0$ is refuted in Lean (`not_staticStrainLinfBound`). The solution-level Prop is not formally refuted: the repo has no local-existence theorem to produce `IsPeriodicMaximalSolution` for u_K, so it remains an unassumed named hypothesis, false modulo classical local existence.

2. **Constant depending on the solution, up to T^*: open, and not enough.** Any smooth solution satisfies $\|S\|_\infty \leq C_T \|\omega\|_\infty$ on each $[0, T]$, $T < T^*$, trivially. Whether C can be taken uniform up to a finite T^* is open. Even if it could, it gives only $M' \leq CM^2$, a lifespan bound, not regularity (cubic-source note, §3).

3. **Positive-time smoothing.** Viscosity regularises the data instantly, so a bound with C depending on ν, $t > 0$ and the energy might hold away from $t = 0$ (the numerics are consistent with this). Near a putative singularity this is exactly what is unknown.

6 Record

Proved (Lean, commit `029a9b9`; standard axioms only): the Fourier identity (`strain00_origin`); $\|\omega_K\|_\infty \leq 36$ (`norm_vort_le`); $\|S_K\|_\infty \geq \frac{1}{\pi^2} \log K$ for all odd K (`originStrain_ge_log`); ¬`StaticStrainLinfBou` (`not_staticStrainLinfBound`), with no caveat. **Classical, not formalized:** $\|\omega_K\|_\infty \leq (6\,\mathrm{Si}(\pi)/\pi)^2$; local existence. **Numerical only:** the $\frac{2}{\pi} \log K$ asymptotic. **Conclusion:** `PeriodicStrainLinfBound` (solution level) is false as stated, modulo classical local existence (not formalized). **Open:** solution-dependent constant up to T^*; not sufficient for regularity in any case.